QuantumRisk v2: Review, Redesign and Strict Antigravity Prompt

Review of the live prototype, a new frontend concept, a clean layout plan, and an ordered build prompt for Antigravity.

1. Review of the Current Prototype

Reviewed from the page content of `/lab` (text and structure; I could not judge the rendered visuals, so check the visual points yourself).

What works

- Clear product framing, a 6-page navigation, and a visible research-prototype disclaimer.
- The right idea: one qubit per clinical feature, with purity shown per qubit.
- Clinical explanation next to each qubit is a good touch for a non-quantum audience.

Problems to fix, most important first

1. **The Lab opens in a dead state.** Every qubit shows theta = 0.00 rad and purity 100%, which means no rotation, no entanglement, nothing to see. The page also shows "Loading quantum state..." and "Evaluating Risk Profile" with no verdict. The first impression should be a live, pre-loaded sample patient with a visible result.
2. **No verdict on the showpiece page.** The risk probability, risk level, and quantum-vs-classical comparison are not on the same screen as the 3D scene. The 3D should serve the answer, not replace it.
3. **Too many controls.** Five view tabs, Auto-Rotate, 2D Grid, a Zoom selector, and six navigation links compete for attention. A first-time viewer does not know what to do.
4. **Misleading claim.** "4 Superconducting Qubits" implies real hardware. It is a simulator. Say "4 simulated qubits (exact statevector)", and show hardware only if you actually run on IBM Quantum.
5. **Over-claiming clinical text.** Phrases like "Confirms anatomical stenosis" and "Directly reflects subendocardial ischemia" read as diagnoses. Soften to "is associated with" or "is a marker of".
6. **Dense, static layout.** Long explanatory blocks sit below the scene; the user has to scroll to connect text to the 3D.
7. **Missing credibility signals:** no AUROC or model card on the page, no uncertainty (shot noise or confidence interval), no sensitivity-first threshold explanation.

Improvements to add (ranked by impact)

#	Improvement	Why it impresses
1	Pre-loaded sample patients (Low, Medium, High) with one click	Instant wow, no empty state
2	Verdict panel: probability, level, confidence interval, top 3 drivers	Answers the question first
3	**What-if sliders** (change cholesterol, watch the circuit and risk update live)	Shows the quantum model reacting
4	**Counterfactual**: smallest change that moves High to Medium	Clinically meaningful and rare in student projects
5	Uncertainty from shot noise (shots slider, error bars)	Shows quantum-specific thinking
6	Shareable URL state (patient values encoded in the link)	Easy demoing in the viva
7	Export a one-page PDF "risk report" with the circuit and benchmark	Tangible deliverable
8	Optional: one real IBM Quantum hardware run shown as a static comparison card	Strongest credibility if you can do it
9	Accessibility: reduced motion, keyboard navigation, colour-blind-safe risk palette, WebGL fallback	Professional polish
10	Performance: lazy-load 3D, instanced meshes, 60fps target	Smooth demo

2. New Frontend Concept: "The Verdict Chamber"

Instead of an observatory with tabs, the new design is a **cinematic, scroll-driven story** with one pinned 3D canvas that changes through five acts, ending in a clear verdict. The user never chooses a tab; they scroll (or press Next), and the 3D transforms.

Core metaphor: a patient's data is *interrogated by a quantum jury*. Each qubit is a "juror" bound to one biomarker. They interact (entangle), then deliver a collective verdict.

Visual identity (deliberately not neon-cyber)

- **Palette:** ink black `#0B0D12`, warm bone `#EDE7DB`, one signal amber `#FFB347`. Risk colours: Low sage `#7FB59A`, Medium amber `#FFB347`, High vermilion `#E5484D`. Max one accent glow at a time.

- **Type:** display serif (Fraunces or Instrument Serif) for headlines, Inter for UI, JetBrains Mono for numbers and quantum notation.
- **Material:** frosted glass panels with hairline borders, soft grain overlay, thin-line 3D (wireframe spheres with glowing arrows), not heavy bloom.
- **Motion:** slow, weighty easing (cubic-bezier 0.22, 1, 0.36, 1); one hero moment per act; nothing loops forever except a faint idle drift.

The five acts (pinned 3D canvas on the right, narrative on the left)

Act	Name	3D scene	Narration headline
1	The Patient	Four floating biomarker tokens (cards with value dials) hover above a faint torso outline	"Four numbers describe this patient."
2	Encoding	Each token dissolves into its own Bloch sphere; the arrow rotates to the encoded angle	"Each number becomes a quantum spin."
3	Entanglement	Light threads connect the spheres; arrows shrink as purity drops; a small circuit strip draws itself below	"The features start to depend on each other."
4	Measurement	The 16 basis states rise as thin pillars; measurement collapses them, pillars split into low-risk (sage) and high-risk (vermilion) halves	"The state collapses into an answer."
5	The Verdict	Pillars merge into a single ring gauge; probability counts up; the quantum vs classical comparison slides in	"Risk: 73%, High."

After Act 5 the page switches to **Explore mode**: the scroll story unpins and the user gets what-if sliders, counterfactual, noise and shots controls, and the benchmark charts, all wired to the same scene.

3. Clean Layout Plan

3.1 Desktop (1440 wide)

```
+-------------------------------------------------------------------------+
| QuantumRisk   [Story | Explore | Benchmarks | Ethics]      sample v
Share|  64px top bar
+------------------------+--------------------------------------+-----------+
|  LEFT RAIL 360px       |  STAGE (pinned 3D canvas, fluid)    | RIGHT 300 |
|  Act number + title    |                                      |  VERDICT  |
|  1-2 line narration    |   Bloch spheres / beams / pillars   |  ring     |
|  Act-specific control  |                                      |  73% High |
|  (e.g. what-if sliders)|                                      |  CI +-4%  |
|  Patient dossier       |                                      |  Drivers  |
|  (4 biomarkers)        |                                      |  vs Models|
+------------------------+--------------------------------------+-----------+
|  PROGRESS BAR: Act 1 --- Act 2 --- Act 3 --- Act 4 --- Act 5   [Prev]
[Next]|  56px
+-------------------------------------------------------------------------+
```

- **One primary action per act**; secondary controls collapse into a single "Settings" popover (rotate, 2D view, shots, noise).
- The **right rail verdict is always visible** from Act 1, greyed and "computing" until Act 5 reveals it.
- Narration is at most two lines; deeper explanation is in an expandable "Why?" drawer.

3.2 Tablet and mobile

- Tablet (768): left rail becomes a bottom sheet; verdict becomes a top chip.
- Mobile (390): the canvas takes the top 55% of the screen, narration card below, act dots at the bottom, verdict as a sticky pill. Free-orbit is off by default; swipe changes acts.
- All 3D has a 2D fallback (SVG Bloch discs and a bar chart) when WebGL is unavailable or `prefers-reduced-motion` is set.

3.3 Page map (reduced from 6 to 4)

1. **/** Landing: one-sentence pitch, hero loop, "Start the demo" and sample patient picker.
2. **/lab** Story then Explore (the showpiece).
3. **/benchmarks** Model comparison, noise and scaling, one scrollable page with anchored sections.
4. **/about** Method, datasets, limits, ethics, model card.

The patient-intake form becomes a slide-over panel inside `/lab` ("Edit patient"), and Training Theatre becomes a section in Benchmarks.

4. Strict Antigravity Prompt (follow in order)

How to use it: paste **Block 0** once as the project rules, then paste **one step at a time**. Do not move to the next step until the current step's acceptance check passes and you have reviewed the Artifact (screenshot or recording). This keeps Flash-class models from drifting.

Block 0: Rules (paste first, keep pinned)

```
You are upgrading the existing QuantumRisk frontend (Next.js 14, TypeScript,
Tailwind, React Three Fiber, drei, framer-motion, Zustand, Recharts).
The FastAPI backend and its /predict JSON contract MUST NOT change.

STRICT RULES
1. Work in the numbered steps I give you, in order. Do ONE step per message.
2. Before coding each step, write a short plan Artifact. After coding,
produce
   a verification Artifact (screenshot or recording, plus lint and type-check
   results). Then STOP and wait for my approval.
3. Never refactor or touch files outside the step's listed scope.
4. No placeholder text, no lorem ipsum, no "coming soon". If data is missing,
   use the fixture in /frontend/lib/fixtures.
5. Every screen must have: loading state, error state, empty state
   (empty state = pre-loaded sample patient, never zeros), and mobile layout.
6. Respect prefers-reduced-motion. Provide a 2D fallback when WebGL fails.
7. Performance budget: 60fps on a laptop, first load JS under 250 KB for
   non-3D routes, 3D code lazy-loaded.
8. Wording: say "simulated qubits", never "superconducting qubits". Clinical
   text must say "associated with" or "marker of", never "confirms" or
   "diagnoses". Keep the disclaimer: Research prototype, not a medical
device.
9. Use ONLY the design tokens from Step 1. No new colours or fonts.
10. If a step is ambiguous, ask me one question instead of guessing.
```

Step 1: Design tokens and typography

```
SCOPE: tailwind.config.ts, app/globals.css, lib/tokens.ts, app/layout.tsx.
Create design tokens: ink #0B0D12, bone #EDE7DB, signal amber #FFB347,
risk-low #7FB59A, risk-med #FFB347, risk-high #E5484D, plus 3 surface levels
and hairline border colours. Fonts via next/font: Fraunces (display),
Inter (UI), JetBrains Mono (numbers). Create base components: GlassPanel,
StatNumber (mono, tabular), SectionTitle, Button (primary amber, ghost).
Add a grain overlay (CSS only, 4% opacity).
ACCEPTANCE: a /dev/tokens page showing every token and component; no
colours outside the token list anywhere in the codebase (grep check).
```

Step 2: App shell and routes

```
SCOPE: app/layout.tsx, app/page.tsx, app/lab, app/benchmarks, app/about,
components/shell/*.
Reduce routes to: /, /lab, /benchmarks, /about. Redirect old routes
(/intake, /compare, /noise, /training, /ethics) to the correct new route.
Build TopBar (64px, logo, nav, sample selector, Share button) and Footer
with the disclaimer. Responsive nav collapses to a menu on mobile.
ACCEPTANCE: all four routes render with correct shell at 1440, 768, 390
widths.
```

Step 3: Data layer and fixtures

```
SCOPE: lib/api.ts, lib/types.ts, lib/store.ts, lib/fixtures/*.
Type the /predict response exactly as in the backend contract. Create three
fixtures (low, medium, high risk patients) with realistic qubit_states,
entanglement matrix, 16 state_probabilities, circuit, explanation and
baselines. Zustand store: patient, result, status, selectedSample, shots,
noiseLevel, act (1-5), mode (story|explore). On first load select the
medium sample and populate the store from its fixture, then call the real
API in the background and swap in the result if it succeeds.
Add URL-state: patient values encoded in query string for Share.
ACCEPTANCE: unit tests for the store; switching sample updates the store;
no page ever shows theta = 0 or purity = 100% for all qubits.
```

Step 4: Lab layout (no 3D yet)

```
SCOPE: app/lab/*, components/lab/Layout, LeftRail, RightRail, ProgressBar.
Build the 3-column layout from the plan: left rail 360px (act title, two-line
narration, act control slot, patient dossier), centre stage placeholder
(dark panel with a loading skeleton), right rail 300px (VerdictRing,
probability with confidence interval, top 3 drivers, quantum vs classical
mini-bars). Bottom ProgressBar with 5 act dots and Prev/Next. Verdict
is visible from Act 1 as greyed and reveals at Act 5. Mobile: stacked layout
with sticky verdict pill and bottom act dots.
ACCEPTANCE: layout correct at 1440, 768, 390; keyboard Left/Right changes
acts.
```

Step 5: 3D foundation

```
SCOPE: components/three/Scene.tsx, CameraRig.tsx, Effects.tsx.
Lazy-loaded R3F canvas with ink background, subtle fog, soft key light,
light postprocessing (very low bloom only). CameraRig has one preset camera
position per act with smooth damped transitions (about 1.2 s). OrbitControls
only in Explore mode. Add WebGL detection and the 2D fallback hook.
ACCEPTANCE: switching act moves the camera smoothly; reduced-motion disables
animation; Lighthouse performance not below 85 on /lab.
```

Step 6: Act 1 and 2 scenes (Patient, Encoding)

```
SCOPE: components/three/BiomarkerToken.tsx, BlochSphere.tsx, acts/Act1.tsx,
acts/Act2.tsx.
Act 1: four floating biomarker tokens with value dials above a faint torso
outline. Act 2: each token dissolves into a thin-line Bloch sphere (axes,
equator ring, glowing arrow). The arrow animates from the pole to the
encoded angle from result.encoded_angles. Arrow length equals purity.
Label each sphere with the biomarker name and theta in mono type.
ACCEPTANCE: recording of Act 1 to Act 2; changing the sample changes arrow
angles; labels never overlap at 390px width.
```

Step 7: Act 3 scene (Entanglement)

```
SCOPE: components/three/EntanglementBeams.tsx, CircuitStrip.tsx,
acts/Act3.tsx.
Draw curved light threads between spheres; thickness and brightness scale
with the entanglement matrix. Arrows shrink smoothly to the real purity.
Below the spheres, a flat CircuitStrip draws the gates from result.circuit
left to right with a travelling pulse. Hovering a thread shows mutual
information value.
ACCEPTANCE: high-risk fixture visibly shows stronger entanglement than the
low-risk fixture; hover tooltip works on touch via tap.
```

Step 7b: Act 4 and 5 scenes (Measurement, Verdict)

```
SCOPE: components/three/ProbabilityPillars.tsx, acts/Act4.tsx, Act5.tsx,
components/lab/VerdictRing.tsx.
Act 4: 16 instanced pillars from state_probabilities; on Next, pillars
collapse and split into low-risk (sage) and high-risk (vermilion) halves by
parity. Act 5: pillars merge into the VerdictRing, probability counts up,
risk level word appears in the display serif, and the right rail reveals
the quantum vs classical comparison.
ACCEPTANCE: pillar heights match the fixture probabilities (test); verdict
ring colour matches risk level for all three samples.
```

Step 8: Explore mode

```
SCOPE: components/lab/ExplorePanel.tsx, WhatIfSliders.tsx,
Counterfactual.tsx,
SettingsPopover.tsx.
After Act 5 unpin the story and enable: what-if sliders for each biomarker
(debounced 250 ms, calls /predict, scene animates between states),
Counterfactual card (calls /explain, shows the smallest change that lowers
the risk level), shots and noise controls with confidence interval error
bars. Collapse rotate, 2D view and zoom into SettingsPopover.
ACCEPTANCE: moving a slider updates sphere angles, pillars and verdict
within 600 ms using the fixture fallback if the API is offline.
```

Step 9: Benchmarks and About pages

```
SCOPE: app/benchmarks/*, app/about/*, components/charts/*.
Benchmarks: one scrollable page with anchored sections: Model comparison
(ROC and AUROC bars for VQC, QSVM, Hybrid, LogReg, SVM, RF, XGBoost,
parameter count and training time table), Noise robustness (line chart),
Scaling (AUROC vs qubits and depth), Training theatre (WebSocket loss
curve). About: method diagram, datasets, limitations, ethics, model card.
All charts use the design tokens and have text alternatives.
ACCEPTANCE: charts render from /metrics or fixtures; keyboard-accessible.
```

Step 10: Polish, export and QA

```
SCOPE: whole frontend, but only fixes and the items below.
Add PDF export of a one-page risk report (verdict, circuit, drivers,
benchmark, disclaimer). Add Share link copy. Run: type-check, lint,
Lighthouse (performance, accessibility 95 plus), test at 1440, 768, 390,
test reduced-motion and no-WebGL. Fix all findings. Produce a final
verification Artifact with screenshots of every route and a 60 second
recording of the full story flow.
ACCEPTANCE: zero console errors; all rules in Block 0 verified in a
checklist.
```

Step 11: Deploy

```
SCOPE: vercel config and env only.
Set NEXT_PUBLIC_API_URL, redeploy, verify the live URL against the checklist
from Step 10, and report any difference between local and production.
```

5. Your Review Checklist After Each Step

- Is the sample patient loaded and visibly non-zero?
- Does the verdict match the fixture risk level?
- Any colour or font outside the tokens?
- Does it work at 390 px and with reduced motion?
- Did the agent touch files outside the step scope?

If any answer is wrong, reply "Fix only: <issue>. Do not change anything else." and wait for the next Artifact.

6. Optional Stretch Ideas (only after Step 11)

- One real IBM Quantum run card (job id, backend name, measured vs simulated probabilities).
- Second condition (Diabetes) with the same story flow and a condition switch.
- A "Judge mode" button that auto-plays the 3-minute demo script with captions.